

Cyber Security Task 01_Anju: Personal Security Audit & Cyber Awareness Assessment

Part D: Cyber Threat Research

1. Phishing

Phishing is a cyber attack where attackers try to trick people into sharing sensitive information such as passwords, OTPs, or bank details. They usually do this through fake emails, messages, or websites that look genuine.

Example - A person receives an email that appears to be from their bank asking them to click a link and update their account details. The link leads to a fake website that steals their login credentials.

Prevention Methods

- Do not click on suspicious links or attachments.
- Verify the sender before responding to emails.
- Check website URLs carefully.
- Enable Multi-Factor Authentication (MFA).

2. Malware

Definition

Malware is malicious software designed to damage, disrupt, or gain unauthorized access to a computer system. Examples include viruses, worms, and spyware.

Real-World Example

A user downloads software from an untrusted website, and it installs a virus that slows down the computer and steals personal information.

Prevention Methods

- Install trusted antivirus software.
- Download files only from reliable sources.
- Keep the operating system updated.
- Avoid opening suspicious attachments.

3. Ransomware

Definition

Ransomware is a type of malware that encrypts files and demands payment to restore access to them. It can make important files completely inaccessible.

Real-World Example

An organization opens a malicious email attachment, and all its files become encrypted. The attackers demand money to provide the decryption key.

Prevention Methods

- Regularly back up important data.
- Avoid opening suspicious emails and attachments.
- Keep software and systems updated.
- Use antivirus and endpoint security solutions.

4. Social Engineering

Definition

Social engineering is an attack where criminals manipulate people into revealing confidential information or performing actions that compromise security.

Real-World Example

An attacker calls an employee pretending to be an IT administrator and asks for their login credentials to "fix a problem."

Prevention Methods

- Never share passwords or OTPs.
- Verify the identity of anyone requesting sensitive information.
- Be cautious of urgent or unusual requests.
- Attend security awareness training.

5. Data Breach

Definition

A data breach occurs when confidential information is accessed, stolen, or exposed without permission. It can involve personal, financial, or business data.

Real-World Example

A company's database is compromised, and customer names, email addresses, and passwords are leaked online.

Prevention Methods

- Use strong and unique passwords.
- Enable Multi-Factor Authentication (MFA).
- Encrypt sensitive information.
- Limit access to important data and monitor systems regularly.